Description

Image





Caption

1. Steel frame building under construction. 2. Construction steelwork. Images used under license from Shutterstock.com

The material

Think of steel and you think of railroads, oilrigs, tankers, and skyscrapers. And what you are thinking of is not just steel, it is carbon steel. That is the metal that made them possible - nothing else is the same time so strong, so tough, so easily formed - and so cheap. Carbon steels are alloys of iron with carbon and, often a little manganese, nickel, and silicon. Low carbon or "mild" steels have the least carbon - less than 0.3%. They are relatively soft, easily rolled to plate, I-sections or rod (for reinforcing concrete) and are the cheapest of all structural metals - it is these that are used on a huge scale for reinforcement, for steel-framed buildings, ship plate and the like.

Compositional summary

Fe/0.02 - 0.3%C

General properties

Density		7.8e3	-	7.82e3	kg/m^3
Price	*	0.918	-	1.02	GBP/kg
Date first used		1610			

Mechanical properties

Young's modulus		200	-	220	GPa
Shear modulus		79	-	84	GPa
Bulk modulus		160	-	180	GPa
Poisson's ratio		0.28	-	0.29	
Yield strength (elastic limit)		255	-	355	MPa
Tensile strength		379	-	532	MPa
Compressive strength	*	255	-	355	MPa

Elongation		25	-	45	% strain
Hardness - Vickers		113	-	168	HV
Fatigue strength at 10^7 cycles	*	203	-	278	MPa
Fracture toughness	*	41.6	-	79	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	9.7e-4	-	0.0014	

Thermal properties

Melting point		1.48e3	-	1.53e3	°C
Maximum service temperature	*	340	-	357	°C
Minimum service temperature	*	-68.2	-	-38.2	°C
Thermal conductor or insulator?		Good conductor			
Thermal conductivity		49.8	-	54.2	W/m.°C
Specific heat capacity		460	-	505	J/kg.°C
Thermal expansion coefficient		11.5	-	13	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good conductor			
Electrical resistivity		15	-	20	µohm.cm

Optical properties

Transparency		Opaque			

Critical Materials Risk

High critical material risk?		No			

Processability

Castability		3			
Formability		4	-	5	
Machinability		3	-	4	
Weldability		5			
Solder/brazability		5			

Durability: water and aqueous solutions

Water (fresh)		Acceptable			
Water (salt)		Limited use			
Soils, acidic (peat)		Acceptable			
Soils, alkaline (clay)		Acceptable			
Wine		Unacceptable			

Durability: acids

Acetic acid (10%)	Limited use
Acetic acid (glacial)	Unacceptable
Citric acid (10%)	Unacceptable
Hydrochloric acid (10%)	Unacceptable
Hydrochloric acid (36%)	Unacceptable
Hydrofluoric acid (40%)	Unacceptable
Nitric acid (10%)	Unacceptable
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Unacceptable
Phosphoric acid (85%)	Unacceptable
Sulfuric acid (10%)	Unacceptable
Sulfuric acid (70%)	Unacceptable

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Acceptable

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Excellent
Crude oil	Excellent
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Limited use
Acetone	Excellent
Ethyl alcohol (ethanol)	Acceptable
Ethylene glycol	Acceptable
Formaldehyde (40%)	Unacceptable

Glycerol	Excellent
Methyl alcohol (methanol)	Acceptable

Durability: halogens and gases

Chlorine gas (dry)	Acceptable
Fluorine (gas)	Excellent
O2 (oxygen gas)	Limited use
Sulfur dioxide (gas)	Acceptable

Durability: built environments

Industrial atmosphere	Limited use
Rural atmosphere	Acceptable
Marine atmosphere	Limited use
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Excellent
Tolerance up to 450 C (842 F)	Acceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	2.23e9	tonne/yr
Reserves, principal component	1.59e11	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 2.24	-	2.46	kg/kg
Embodied energy (virgin grade production)	* 30.4	-	33.6	MJ/kg
Water usage	* 43.2	-	47.7	l/kg

Material processing: energy

Metal casting, embodied energy	* 48.4	-	55.1	MJ/kg
Metal extrusion, hot, embodied energy	* 13.6	-	18	MJ/kg
Metal rolling and forging, embodied energy	* 7.27	-	9.57	MJ/kg
Wire drawing, embodied energy	* 4.38	-	4.83	MJ/kg

Metal powder forming, embodied energy	* 69.3	-	84.8	MJ/kg
Chemical vapor deposition (CVD), embodied energy	* 1e4	-	1.13e4	MJ/kg
Machining, coarse, embodied energy	* 16.5	-	18.7	MJ/kg
Machining, fine, embodied energy	* 20.3	-	23.8	MJ/kg
Machining, grinding, embodied energy	* 24.1	-	28.9	MJ/kg
Machining, laser, CO2, embodied energy	* 1.19	-	1.34	MJ/m

Material processing: climate change

Metal casting, climate change (CO2-eq)	* 3.49	-	3.99	kg/kg
Metal extrusion, hot, climate change (CO2-eq)	* 0.888	-	1.18	kg/kg
Metal rolling and forging, climate change (CO2-eq)	* 0.476	-	0.627	kg/kg
Wire drawing, climate change (CO2-eq)	* 0.286	-	0.317	kg/kg
Metal powder forming, climate change (CO2-eq)	* 5.66	-	6.92	kg/kg
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 823	-	926	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.08	-	1.22	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.33	-	1.56	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.58	-	1.9	kg/kg
Machining, laser, CO2, climate change (CO2-eq)	* 0.098	-	0.11	kg/m

Recycling and end of life

Recycle	✓			
Functional recycle	✓			
Climate change (CO2-eq), recycling	* 0.605	-	0.664	kg/kg
Embodied energy, recycling	* 9.42	-	10.4	MJ/kg
Recycle fraction in current supply	41			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

The production energy of steel is comparatively low - per unit weight, about a half that of polymers; per unit volume, though, twice as much. Carbon steels are easy to recycle, and the energy to do so is small.

Supporting information

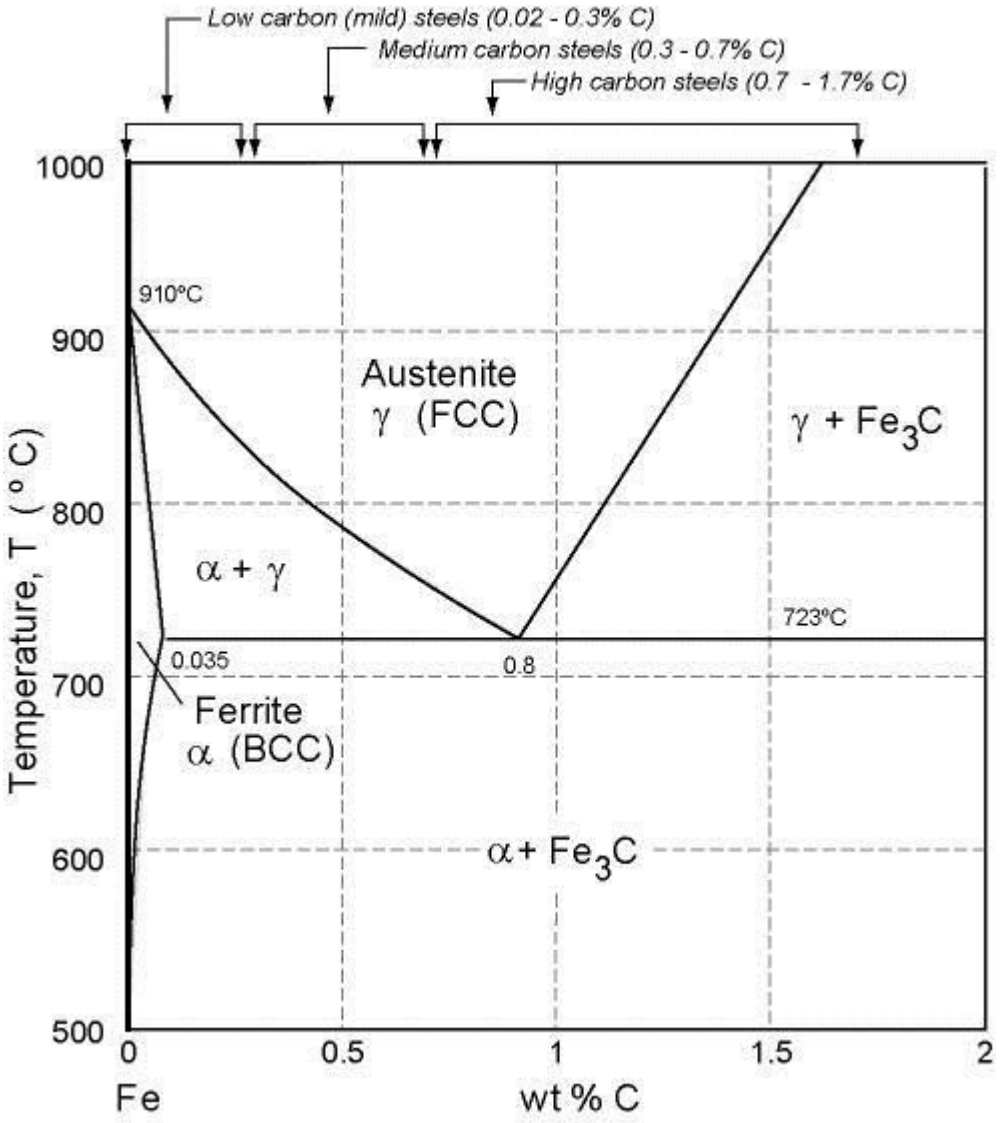
Design guidelines

Hardenability measures the degree to which it can be hardened in thick sections. Low carbon steels have too little carbon to harden much, and have poor hardenability - additional alloying elements are used to increase it (see Low alloy steels).

Technical notes

The two standard classifications for steels, the AISI and the SAE standards, have now been merged. In the SAE-AISI system, each steel has a four-digit code. The first two digits indicate the major alloying elements. The second two give the amount of carbon, in hundredths of a percent. Thus the plain carbon steels have designations starting 10xx, 11xx, 12xx or 14xxx, depending on how much manganese, sulfur and phosphorus they contain. The common low-carbon steels have the designations 1015,1020, 1022, 1117,1118; the common medium carbon steels are 1030,1040, 1050, 1060, 1137, 1141, 1144 and 1340; the common high alloy steels are 1080and 1095.

Phase diagram



Phase diagram description

Low carbon steels are alloys of iron (Fe) with 0.02 - 0.3% carbon (C), for which this is the phase diagram.

Typical uses

Low carbon steels are used so widely that no list would be complete. Reinforcement of concrete, steel sections for construction, sheet for roofing, car body panels, cans and pressed sheet products give an idea of the scope.

Links

ProcessUniverse

Producers

Reference

Values marked * are estimates.
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